6. Vacuum Cleaner Problem

Aim:

To write a Python program to implement the Vacuum Cleaner problem using an intelligent agent.

Algorithm:

1. Start the program.

2. Define two rooms, A and B.

3. Assign the status of each room as clean or dirty.

4. Initialize the vacuum cleaner position.

5. Check the current room status.

6. If dirty, clean the room.

7. If clean, move to the other room.

8. Repeat until both rooms are clean.

9. Display the final status.

10. Stop.

CODE

```
rooms = {
    "A": "Dirty",
    "B": "Dirty"
}

location = "A"

while "Dirty" in rooms.values():
    if rooms[location] == "Dirty":
        rooms[location] = "Clean"
        print("Cleaning", location)
    else:
        location = "B" if location == "A" else "A"
        print("Moving to", location)

print("Final State:", rooms)
```

Output:

```
Cleaning A
Moving to B
Cleaning B
Final State: {'A': 'Clean', 'B': 'Clean'}
```

Result:

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